A decentralized neutrosophic decision layer for global-state freshness in microservices architectures

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ABSTRACT

Microservices decide over global state, yet each owns its data and the only cross-service signal is a cache that may hold uncommitted values. A boolean fresh/stale flag cannot express a value held only in cache: neither committed nor absent, but *indeterminate*. We develop a decentralized decision layer where each item carries a single-valued neutrosophic triple (T,I,F) for persistence status; peers fuse their views with weighted operators and a deneutrosophy score and act or abstain without central authority. The layer sits above the consistency substrate and is not a consensus protocol. We prove one-round termination and conservative-safety/optimistic-liveness bounds (generalized to a graded encoding); a Byzantine analysis separates gate from value robustness; and operator, score, and weight audits confirm it is not a design artefact. Against consistency baselines it occupies a Pareto interior: at a 10% failure rate it cuts the stale-decision rate by 16.7 points versus last-writer-wins, stays 88% available under partition where a primary drops to 45%, matches a boolean quorum's correctness at twice its availability, and generalizes to a distinct feature-store domain. We claim no superiority over consensus; the contribution is a transparent, reproducible trade-off whose every reported number regenerates from committed code, a content-hashed calibration, and fixed seeds.

Keywords: microservices; neutrosophic sets; data freshness; eventual consistency; Byzantine fault tolerance; decentralized decision-making.

1 Introduction

Microservices architectures decompose an application into sovereign services, each owning its data and exposing it only through APIs[1]. This sovereignty is precisely what makes *global-state* decisions hard: computing a personalized price, deciding whether an order is fraudulent, or determining whether a freshly cloned replica may begin serving all require data that lives in other services and may be only transiently cached. Strong consistency removes the uncertainty but reintroduces a coordination point that contradicts the architecture's goals and stalls under partition[2,3]; eventual consistency keeps services available but exposes them to *dirty reads*—acting on a cached value that was never committed and will not become the truth[4].

The difficulty is that a boolean "fresh/stale" flag collapses two qualitatively different states into one: a cached-but-unconfirmed value is treated identically to an absent value, even though it usually holds the latest write. What a boolean bit discards is *indeterminacy*. Annotating a data item with a single-valued neutrosophic triple (T,I,F)—reading persisted, cached, and absent as truth, indeterminacy, and falsity—retains the unconfirmed-but-present state, so a service can make a calibrated act-or-abstain decision rather than a brittle binary one. The neutrosophic encoding of storage status was introduced conceptually in[5]; this paper develops it into a concrete decentralized decision layer with an explicit protocol, formal guarantees, and a controlled empirical evaluation.

Contributions.

1. A decentralized (T,I,F)-aggregating decision protocol for global-state freshness: each peer encodes its storage status as a single-valued neutrosophic number, peers fuse their views with weighted neutrosophic operators and a deneutrosophy score (a crisp confidence derived from the triple), and a service acts or abstains with no central authority. The layer is a per-item freshness decision above the consistency substrate, not a replication or consensus protocol; its directly comparable control is a boolean quorum (Section 4).

2. Formal guarantees for the protocol: one-round termination and partition tolerance, a conservative-safety and an optimistic-liveness bound that bracket the act/abstain gate, and Byzantine gate-resistance under robust aggregation (Section 4.5).

3. A two-axis audit that tests whether the result is an artefact of design choices: an aggregation-operator panel (Einstein, Hamacher, Dombi, Aczél–Alsina, Bonferroni) over a graded confidence encoding, a deneutrosophy-score panel, and an equal-budget metaheuristic tuning of the threshold and peer weights (Section 4.4).

4. A robustness study under an adversarial (Byzantine) failure model with median-, trimmed-mean-, and Krum-based robust aggregation, and a freshness-SLO baseline that makes the comparison to adaptive-consistency systems empirical rather than positional (Sections 4.4, 2).

5. A reproducible evaluation on an e-commerce testbed against a panel of consistency baselines (strong, eventual, quorum, and partial-quorum) with a privileged ground-truth oracle and held-out threshold calibration. The design uses ≥ 50 repetitions per cell across five seeds, Holm correction for multiple comparisons, and bootstrap confidence intervals; every reported number can be regenerated from committed code, a content-hashed calibration file, and recorded seeds (Section 2).

The operator, score, and weight audits are not the core novelty but establish generality: the freshness result is a property of the (T,I,F) representation and its gate, not of any single operator, score, or weight setting.

Contemporary relevance. The problem of deciding whether locally-held, possibly-stale shared state is trustworthy enough to act on is surfacing across four 2024–2026 research currents: multi-agent and LLM-agent systems that must reason over shared memory and name consistency as a leading open challenge[6]; the edge-cloud continuum and serverless state, in which an actor reads ephemeral or expired state[7]; ML feature stores that operationalize feature-freshness service-level objectives[8,9]; and microservice data-consistency surveys and benchmarks[4,10,11]. Each collapses a graded freshness signal to a boolean or a single probability, whereas the (T,I,F) encoding retains the indeterminacy axis these settings require.

1.1 Related work

Microservices state and consistency. Surveys of microservice data management[1,4] document that per-service sovereignty leaves no global view, and the BAC theorem[2] formalizes the backup–availability–consistency tension specific to microservices. These works frame the problem we target; none provide a freshness-aware decision mechanism.

Consistency models and CRDTs. Causal+[12], probabilistically bounded staleness (PBS)[13], conflict-free replicated data types[14,15], the PACELC trade-off[3], and surveys of non-transactional consistency models[16] give the vocabulary of staleness. PBS is the closest prior art for "probabilistic freshness," but it reasons about quorum probabilities over a *boolean* read, not a three-valued per-peer provenance fused into an act/abstain decision; we therefore include a PBS-style partial-quorum reader as an explicit baseline (Section 4).

Consensus (for positioning). Raft[17], Paxos[18], and Byzantine fault tolerance[19] solve *agreement* on a replicated log. Our layer is orthogonal: it decides, above whatever substrate exists, whether the value a service already holds is fresh enough to act on. We therefore use a Raft-style last-writer-wins reader only as a reference point, never as a competitor for the same job.

Rules engines and DSLs. Production rule systems (Rete[20], Drools[21]) and DSL tooling[22,23] are uniformly *boolean* substrates: a fact is present or not. We reuse ANTLR to parse a JSON rule DSL but make the freshness of each fact a graded, neutrosophic quantity.

Neutrosophic aggregation. Single-valued neutrosophic sets[24,25] and their weighted operators and score functions[26–28] are mature, but applied almost exclusively to offline multi-criteria decision making, including the authors' prior work[29]. To our knowledge no prior work encodes runtime persistence/freshness as a (T,I,F) triple and fuses peer views into a decentralized systems decision. This intersection—neutrosophic aggregation applied to microservice state—is the gap we fill.

Accuracy-aware staleness and adaptive consistency. The closest modern work treats staleness as a first-class, accuracy-aware quantity: RALF schedules feature-store maintenance against a downstream-accuracy "regret"[9], and adaptive-consistency tuners select consistency levels under cost/SLO pressure. These are *centralized* schedulers/tuners; our layer is the *decentralized*, per-item act/abstain dual—each peer fuses (T,I,F) views without a central accuracy oracle. We position RALF as the nearest modern point of comparison and deepen the contrast with PBS (a single staleness probability vs. a fused three-valued provenance).

Reproducible evaluation. We follow the microservices benchmarking tradition[30,31], statistically rigorous performance methodology[32], and reproducibility principles for cloud experiments[33]: multiple repetitions and seeds, effect sizes with confidence intervals, multiple-comparison correction, and a content-hashed, single-aggregator pipeline.

Table 1 positions this work against the prior families. The distinguishing axis is the *freshness signal*: prior readers collapse it to a Boolean (committed or not), a single staleness probability, or a centralized SLO, whereas the layer here carries a fused three-valued (T,I,F) provenance, decentrally, with an explicit abstain option and Byzantine-robust variants.

Table 1. Where the neutrosophic freshness layer sits relative to prior approaches to global-state reads. "Abstain" = can decline to act under uncertainty; "Decentralized" = no coordinating leader.

Approach	Freshness signal	Abstain	Byz.	Decentralized
Strong consistency (Raft/Paxos)[17,18]	committed (serialized)	no	via PBFT	no
Eventual / LWW CRDT[14]	none (serve latest)	no	no	yes
Bounded staleness (PBS)[13]	one probability	no	no	quorum
Consistency SLA (Pileus)[34]	per-read SLO	no	no	no
Feature-freshness scheduler (RALF)[9]	centralized regret	no	no	no
Boolean quorum (control)	1 bit (fresh/stale)	yes	no	yes
This work	(T,I,F) **triple**	**yes**	**median/Krum**	**yes**

2 Results

Across the validated grid (288 cells over five master seeds), 98% of headline cells replicate in sign across all seeds. Supplementary Fig. S1 shows stale-decision rate versus failure-injection rate; Figure 1 the correctness/availability trade-off; Figure 2 availability under partition. Supplementary Table S2 reports S2 in full; Supplementary Table S3 reports S1.

2.1 Signal richness

The indeterminacy axis is heavily populated (mean cache occupancy ≈ 0.4), and the neutrosophic decision differs from a boolean quorum on a substantial fraction of decisions—decisions the boolean flag cannot make without abstaining (Supplementary Fig. S2).

2.2 Correctness and availability under failure

At $\varphi = 0.1$ (S2, no partition), neutro-waa (the optimistic weighted-arithmetic configuration of the layer; Section 4) attains a stale-rate of 0.248 $[0.244, 0.251]$ versus 0.415 for lww-crdt, 0.506 for naive-cache, and 0.349 for the PBS partial-quorum reader (the nearest prior art)—absolute reductions of 16.7, 25.8, and 10.1 percentage points (40%, 51%, and 29% relative), respectively (two-sided paired Wilcoxon signed-rank over $n = 250$ paired repetitions, $\alpha = 0.05$, Holm-corrected $p < 10^{-30}$; Cohen's $d = -4.4$ vs. lww-crdt). In S1 the reduction versus naive-cache reaches 24.9 percentage points (65% relative; 0.137 vs. 0.386) at full availability.

RQ3/RQ4: the availability–correctness trade-off. The strongly-consistent baselines (centralized, raft-lww) achieve zero staleness but lose availability under failure, collapsing to 0.453 under transient partition where neutro-waa remains 0.880 available (Cohen's $d = 12.6$, Holm-corrected $p < 10^{-30}$; Figure 2). Neutro-waa thus occupies the Pareto interior (Figure 1): nothing dominates it—centralized trades availability for its zero staleness, and every full-availability baseline is far staler.

RQ2 vs. the boolean control. Quorum-bool reaches a low stale-rate only by abstaining on most decisions (0.402 availability at $\varphi = 0.1$, S2); neutro-waa attains comparable correctness at 0.942 availability—roughly $2.3\times$. This isolates the contribution: retaining (T,I,F) rather than a boolean flag converts indeterminacy into available, correct decisions.

2.3 Comparison with a single-probability gate

A one-bit quorum is an easy control; the sharper test is a *calibrated single-probability* gate (prob-gate) that collapses the same evidence into one Bayesian freshness posterior and acts when it clears τ. prob-gate is a strong baseline—far above the boolean quorum (availability 0.998 vs. 0.402 at S2, $\varphi = 0.1$)—and is *not* dominated by the neutrosophic layer: on S1 the two are near-identical (0.136 vs. 0.137 stale at full availability), and on S2 prob-gate trades 2.8 percentage points more staleness (0.263 vs. neutro-waa-g's 0.235) for 6.9 points more availability (0.998 vs. 0.929). The result is reported as it stands: (T,I,F) does not beat a calibrated scalar on the headline trade-off, but it is not a mere relabelling either. Calibrated for the asymmetric cost, the single scalar collapses toward "act on almost everything" (0.998 availability), whereas the indeterminacy axis lets the layer *selectively abstain* on genuinely-unconfirmed decisions, and the operator/score/threshold family spans the whole frontier (Figure 3) that one calibrated point cannot. The value of the three-valued encoding is representational—frontier control and an interpretable separation of indeterminacy from falsity—not a uniform accuracy win over a strong scalar.

2.4 Operator, score, and weight robustness

SVNNWGA (the single-valued neutrosophic weighted geometric averaging operator) is the conservative operator: lower stale-rate (e.g. S3 transient at $\varphi = 0.1$: 0.218 vs. 0.288) at much lower availability (0.279 vs. 0.993). The arithmetic (SVNNWAA) and geometric (SVNNWGA) operators thus span an optimistic–conservative frontier, selectable per scenario via τ.

Operator panel (Axis A). Beyond the crisp WAA/WGA pair, a panel of parametric t-norm/t-conorm operators evaluated on the graded encoding (Einstein, Hamacher, Dombi, Aczél–Alsina, Bonferroni; Section 4.4) spreads a tunable correctness/availability frontier at *fixed* τ (Figure 3). At $\varphi = 0.1$ (S2) the operators range from conservative (Aczél–Alsina: stale 0.162 at availability 0.624) through intermediate (Einstein 0.187/0.750; Dombi 0.202/0.842) to optimistic (median and Krum 0.264/1.000), with WAA (0.235/0.929) in between. No single operator dominates—the choice selects a position on the frontier, a second tuning axis beyond τ. On crisp inputs every operator collapses exactly to WAA (a verified null), so the spread is a property of the graded encoding, not of the operators in isolation.

Score robustness (Axis A′). Swapping the deneutrosophy score that maps an aggregated (T, I, F) to the gate—an accuracy, a cosine, and a truth-weighted score in place of the default linear score—shifts a system along the same frontier without changing the qualitative result (S2, $\varphi = 0.1$): stale-rate 0.235 (linear), 0.227 (accuracy), 0.219 (cosine), 0.191 (truth-weighted), at availabilities from 0.929 down to 0.762. The Pareto-interior advantage is thus a property of the (T, I, F) representation and its gate, not of any single score.

Weight and threshold tuning (Axis B). Replacing the hand-fit τ and uniform weights with an equal-budget metaheuristic search over $[\tau, w_1, \ldots, w_R]$—seven optimizers (random, DE, CMA-ES, TPE, PSO, RIME, NiOA) at 120 evaluations each (Supplementary Fig. S8)—gives a two-part result. First, tuning helps only where the cost is asymmetric: on the symmetric scenarios S1 and S3 every optimizer recovers *exactly* the uniform-weights utility (0.667 and 0.292—the hand-fit calibration is already optimal), whereas on the asymmetric-cost S2 the best search improves utility by 10.1% ($-0.079 \to -0.071$) by weighting peers unequally. Second, at equal budget the optimizer choice is largely immaterial: all seven converge to an identical utility on S1 and S3 and to within $\approx$2% on S2, where a strong optimizer (TPE) beats uniform while a poorly matched one (CMA-ES) falls slightly below it. The tuning gain is therefore real but modest and regime-specific, and—consistent with the No-Free-Lunch theorem[35]—not an artefact of choosing a fashionable optimizer.

2.5 Byzantine robustness

We stress the layer with peers that claim *persisted* while serving a fabricated value at an inflated version, and compare the Byzantine-robust aggregators (coordinate-wise median, trimmed mean, Krum) against the averaging operators (Figure 4, S2, $R = 5$). With no adversary the readers behave normally (stale-rate 0.235–0.264). Once a single adversary appears (1 of 5 peers), *all* neutrosophic readers degrade together—median, Krum, trimmed mean, and WAA alike rise to stale-rate 0.718, and to 0.984 at two adversaries; eventual consistency reaches 1.0. The robust aggregators do *not* outperform the averaging operators here, and we do not claim they do. This is informative rather than negative: robust aggregation defends the *gate*—no adversarial minority can fabricate the *confidence* needed to cross τ on a stale value (Lemma 3 bounds the perturbation of the aggregated indeterminacy)—but a freshness layer's dominant exposure is value *selection*, which *value* to return among reachable peers, and coordinate-wise robustness on the (T, I, F) triple does not govern that choice. The interpretation is a scope statement: the mechanism tolerates confidence inflation but not value fabrication, and closing the latter calls for a robust value-quorum, which we leave to future work.

2.6 Latency and throughput

On a real localhost HTTP testbed (Section 4), the strategies separate into three latency tiers that confirm the communication-cost model (Supplementary Fig. S3): a local read (naive-cache, ≈ 0 ms), a single round-trip (centralized, raft-lww, single-peer), a partial quorum (pbs-quorum), and a full R-peer fan-out (neutro-waa/wga, quorum-bool, lww-crdt). At a 5 ms per-hop RTT the neutrosophic layer's median decision latency is in the fan-out tier alongside quorum and CRDT—the measured price of decentralized decision-making, compared to a single hop for the centralized authority. (Absolute values include loopback runtime overhead; the operative cost axis is messages per decision, reported exactly in Supplementary Tables S2 and S3.) Supplementary Fig. S4 reports measured throughput under concurrent load: the local and single-hop strategies sustain the highest decisions/sec, and the fan-out strategies (including ours) the lowest, consistent with their message cost. The same tiers hold on a containerized deployment (five peer containers plus a Redis bus and Postgres under Docker Compose): a single-hop read measures ≈ 9.6 ms median and the R-peer fan-out ≈ 26.4 ms median, confirming the result is not an artefact of the in-process harness.

Decision latency under emulated WAN. We harden the latency claim on the same containerized deployment with a Toxiproxy proxy injecting per-link latency, jitter, and loss (Supplementary Fig. S7; single-host *emulation*—real HTTP over real proxied sockets, not a geo-distributed cluster). The three communication patterns separate cleanly and the gap *grows* with link latency: at a 1 ms LAN profile the median single-hop, majority-wait, and R-peer fan-out decisions are 3.1, 8.3, and 13.1 ms; at an 80 ± 20 ms WAN profile they are 83, 251, and 418 ms (p99 103, 296, 479 ms). The fan-out tail compounds because a decision waits for the slowest of R peers, so jitter and—more sharply—loss inflate it: under 2% packet loss the R-peer fan-out p99 rises to 1,330 ms (a dropped reply bounds the decision at the 1 s deadline) against 1,003 ms for the single hop. The measured

Table 2. Headline paired contrasts (S2, $\varphi = 0.1$): paired Wilcoxon, Holm-corrected within family; Cohen's d from paired differences (250 trials/cell).

Metric	Contrast	Partition	Cohen's d	p_{holm}
stale-rate	neutro-waa vs. lww-crdt	none	-4.42	7.9e-40
stale-rate	neutro-waa vs. lww-crdt	transient	-3.25	7.9e-40
stale-rate	neutro-waa vs. naive-cache	none	-8.10	7.9e-40
stale-rate	neutro-waa vs. naive-cache	transient	-7.44	7.9e-40
stale-rate	neutro-waa vs. quorum-bool	none	-0.48	3.5e-10
stale-rate	neutro-waa vs. quorum-bool	transient	+1.48	1.5e-37
availability	neutro-waa vs. centralized	none	+1.82	7.6e-40
availability	neutro-waa vs. centralized	transient	+12.56	7.2e-40

price of decentralized decision making is thus a fan-out tail that grows with both latency and loss—the operative cost the message-count column abstracts, now quantified on real network paths. Absolute values remain host-bound (shared CPU/NIC); a geo-distributed multi-datacenter study is future work (Section 3).

Sensitivity. The advantage is stable across replica counts $R \in \{3,5,7\}$ and grows with the cache ratio ρ (Supplementary Fig. S6): as more reads are unconfirmed, the gap between the neutrosophic layer and the eventual-consistency baselines widens, because that is exactly the regime in which the indeterminacy axis carries information.

2.7 Scalability and cross-domain generality

Sweeping the replica count $R \in \{2,\ldots,50\}$ at fixed τ (Figure 5), the neutrosophic layer's stale-rate *falls* monotonically with R—neutro-waa from 0.334 at R=2 to 0.099 at R=50, with availability rising from 0.805 to 1.0—at a per-decision message cost that scales linearly (R messages, confirmed exactly in the sweep). More peers supply more evidence, so the aggregate is both more available and more correct. The contrast with the baselines is sharp: eventual consistency *degrades* with scale (lww-crdt stale-rate rises from 0.367 to 0.994 as more replicas diverge), and the boolean quorum's availability stays low throughout (0.28–0.41). The layer therefore exhibits favourable scaling: adding peers improves rather than degrades decision quality.

Cross-domain generality (S4). To test whether the result is specific to e-commerce, S4 instantiates a distinct domain—an ML online feature store (high version-lag and dirty-write rates, served largely from cache, high stale-cost; Section 4). The Pareto-interior structure replicates: at $\varphi = 0.1$ neutro-waa attains a stale-rate of 0.402 at 0.953 availability, fresher than eventual consistency (lww-crdt 0.427 at 1.0) and the single-probability gate (prob-gate 0.412 at 0.998), and far more available than the boolean quorum (0.452 at only 0.451); the graded operator trades more abstention (0.626 availability) for a markedly lower stale-rate (0.327). The regime is harder overall—every reader is staler than on the e-commerce scenarios—and the margins over the strong *scalar* baselines are modest, consistent with the single-probability finding; but the qualitative advantage of retaining the (T,I,F) signal carries to a second, structurally different domain.

S3: clone catch-up. A freshly cloned replica boots with every key absent and replays the logs-as-DSL stream one event at a time (Supplementary Fig. S5). Recovered state grows linearly with events replayed; under the conservative SVNNWGA the clone drags cluster consensus to reject until it has nearly caught up (95% acceptance after 48 of 50 events), whereas under the optimistic SVNNWAA the established peers carry every decision at 100% acceptance throughout the catch-up window. The operator choice thus governs the availability-vs-strictness behavior precisely during the "White Friday" load-spike scenario (S3).

Significance. Table 2 lists the headline paired contrasts with Holm-corrected p-values and effect sizes; all are significant after correction, with large effect sizes for the comparisons against eventual consistency and the availability gain under partition.

3 Discussion

The neutrosophic layer is worth its cost where decisions must continue under failure and where acting on a never-committed cached value is harmful—exactly the regime of fraud detection (S2) and clone-readiness (S3). It is *not* worth it when strong consistency is mandatory (it does not reach zero staleness) or when a cheap naive read is correct enough (low φ, low dirty-write rates). The operator and threshold give an operator-tunable dial: optimistic WAA with a high τ for high-cost decisions, conservative WGA where abstaining is acceptable. The indeterminacy axis is the source of the gain: by refusing to collapse "cached-unconfirmed" into "stale," the layer keeps making correct decisions where a boolean quorum must abstain.

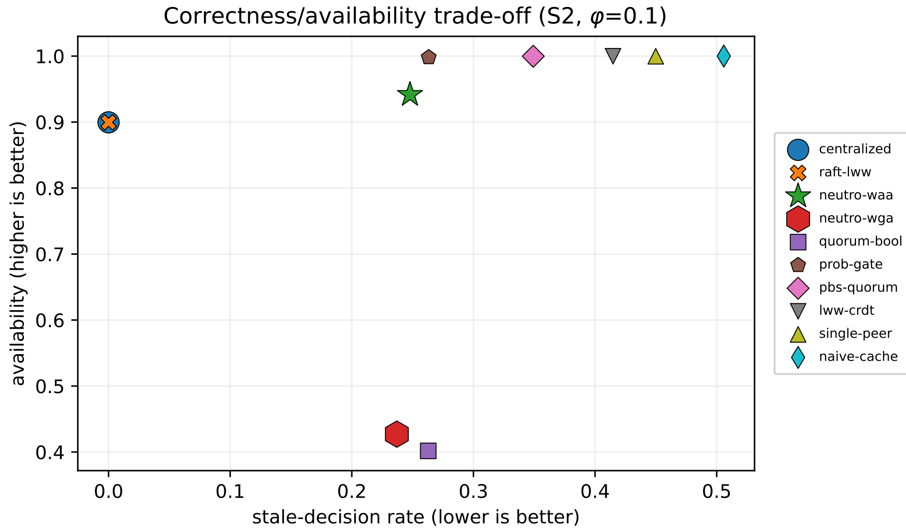

Figure 1. Correctness/availability trade-off (S2, $\varphi = 0.1$). Up-and-left is better; neutro-waa is on the Pareto frontier. Each system has a distinct marker; the strongly-consistent baselines centralized and raft-lww coincide at $(0, 0.90)$ (circle behind cross).

The DSL front-end is interchangeable. State and rule messages are parsed by an ANTLR4 grammar, but nothing in the decision layer depends on that choice: the parser only turns a message into a typed event for the rules engine, and the neutrosophic aggregation, gate, and protocol operate on the resulting (T, I, F) views. ANTLR4 remains a standard generator—with *antlr-ng* a current drop-in successor—and the front-end could equally be a tree-sitter grammar (incremental, linear-time), a Langium/LSP workbench, or a Lark grammar without altering any result in this paper. The contribution is the decision mechanism, not the parser; we keep ANTLR for its mature multi-target tooling.

Threats to validity.

Construct. The layer is a freshness-decision mechanism, not a consensus protocol; comparisons to Raft/Paxos as if they did the same job would be invalid, so we use them only as reference points and make quorum-of-booleans the controlled comparison. "Stale" is defined against an oracle's causal-observation model; we unit-test the oracle.

Internal. τ and weights could be tuned to favor our arm; we fit them on held-out calibration seeds and freeze them. The paired seed design gives every strategy identical failure and arrival streams.

External. The correctness/availability evaluation is simulation-based on synthetic e-commerce workloads (sensitivity to R and ρ is probed in Supplementary Fig. S6); how faithfully these map to production microservice traces is untested, and replaying real traces is future work. Decision latency is measured on a *real* HTTP testbed—in-process and on a single-host Docker Compose deployment—which confirms the communication tiers, but the *absolute* values include local runtime overhead and a shared host NIC; we therefore read the latency result as the *relative* ordering of the tiers, not as milliseconds. An *emulated*-WAN study under injected per-link latency, jitter, and loss (Supplementary Fig. S7) confirms the tiers and tail behaviour on real network paths, but it is single-host; a *geo-distributed* multi-datacenter study remains future work. Three further limitations bound the scope: (i) the protocol assumes a *static* reachable peer set per decision—service churn, elastic scale-out, and dynamic membership are not modeled and would need a membership/discovery layer beneath the gate; (ii) the DSL event bus adds an operational hop and serialization cost that a production deployment must budget; and (iii) adversarial behavior is studied in simulation (Figure 4) under a value-fabrication model, so a real-testbed adversarial study and the robust value-quorum that Proposition 3 calls for remain future work.

Conclusion. Latency-like quantities are heavy-tailed, so we use nonparametric paired tests and report effect sizes with bootstrap CIs; multiple comparisons are Holm-corrected within a-priori families; cross-seed non-replication (2% of cells) is flagged and excluded from headline claims.

Conclusion. We developed a decentralized neutrosophic decision layer for microservice global-state freshness—a protocol, formal safety and liveness guarantees, and a reproducible evaluation—and showed that encoding persistence as a (T, I, F)

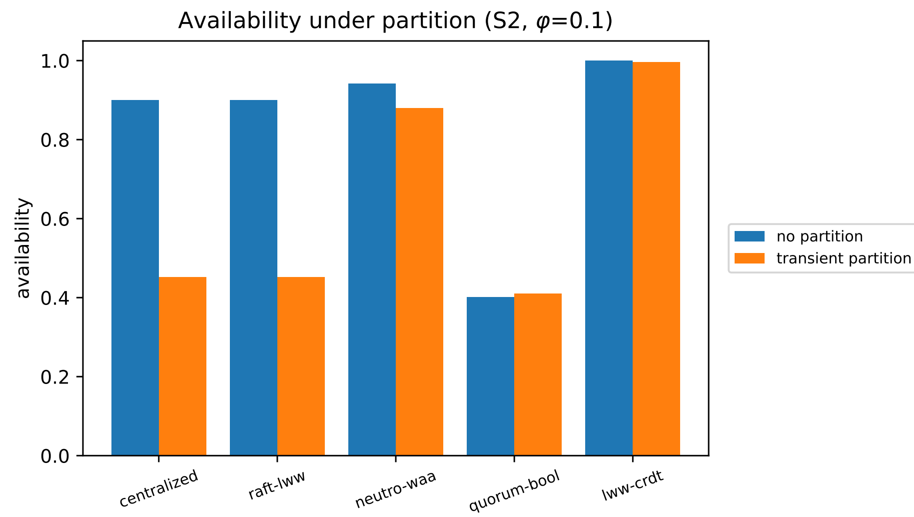

Figure 2. Availability without vs. with transient partition (S2, $\varphi = 0.1$). The strongly-consistent baselines stall; the neutrosophic layer stays available.

triple and fusing peer views with weighted neutrosophic operators yields a decision layer on the Pareto interior between strong and eventual consistency: markedly fewer stale decisions than eventual-consistency baselines, sustained availability under partition where a strongly-consistent primary stalls, and a clear advantage over a boolean quorum that isolates the value of the indeterminacy axis. We claim no superiority over consensus protocols; the contribution is a transparent, calibrated, and fully reproducible trade-off.

Future work. The clearest next step is a Byzantine-robust *value* quorum beneath the gate—Proposition 3 shows it is needed to close the value-fabrication gap that gate robustness alone leaves open. Further directions: hardening the deployment study toward geo-distributed, multi-datacenter operation; replaying production microservice traces in place of synthetic workloads; learned, reputation-weighted peer weights; and dynamic membership and service churn beneath the gate.

4 Methods

4.1 Neutrosophic preliminaries

A single-valued neutrosophic number (SVNN) is a triple $z = (T, I, F)$ with $T, I, F \in [0,1]$ denoting independent truth-, indeterminacy-, and falsity-membership[24]. We map a data item's storage status to canonical SVNNs: *persisted* $= (1,0,0)$, *cached* $= (0,1,0)$, and *absent* $= (0,0,1)$. An unseen value reads as falsity rather than indeterminacy, which matters because a missing or partitioned peer contributes the absent state; encoding absent as $(0,0,1)$ rather than $(0,1,0)$[5] keeps it distinct from *cached*.

Given peer views $z_1, \ldots, z_n$ with normalized weights w_j, the SVNN weighted arithmetic and geometric averaging operators are

$$\mathrm{SVNNWAA} = \Big(1 - \textstyle\prod_j (1 - T_j)^{w_j},\ \prod_j I_j^{w_j},\ \prod_j F_j^{w_j}\Big), \tag{1}$$

$$\mathrm{SVNNWGA} = \Big(\textstyle\prod_j T_j^{w_j},\ 1 - \prod_j (1 - I_j)^{w_j},\ 1 - \prod_j (1 - F_j)^{w_j}\Big). \tag{2}$$

We evaluate the products in log space for numerical stability at large n. The deneutrosophy score $s(z) = (2 + T - I - F)/3 \in [0,1]$[26] collapses an aggregate to a crisp confidence; a service *acts* iff $s \geq \tau$ for a calibrated threshold τ. SVNNWAA is optimistic: a single persisted peer ($T = 1$) carries the decision. SVNNWGA is conservative: a single uncommitted or absent peer ($T = 0$) vetoes the aggregate truth.

4.2 Architecture, DSL, and rules engine

The testbed uses a conventional e-commerce decomposition (products, orders, users, front-end); Figure 6 shows the overall decision layer. Each service holds a two-tier (cache/database) store that tags every value with its (T, I, F) status, and services exchange JSON *state* and *rule* messages over a lightweight event bus, never reading each other's stores directly. Both message

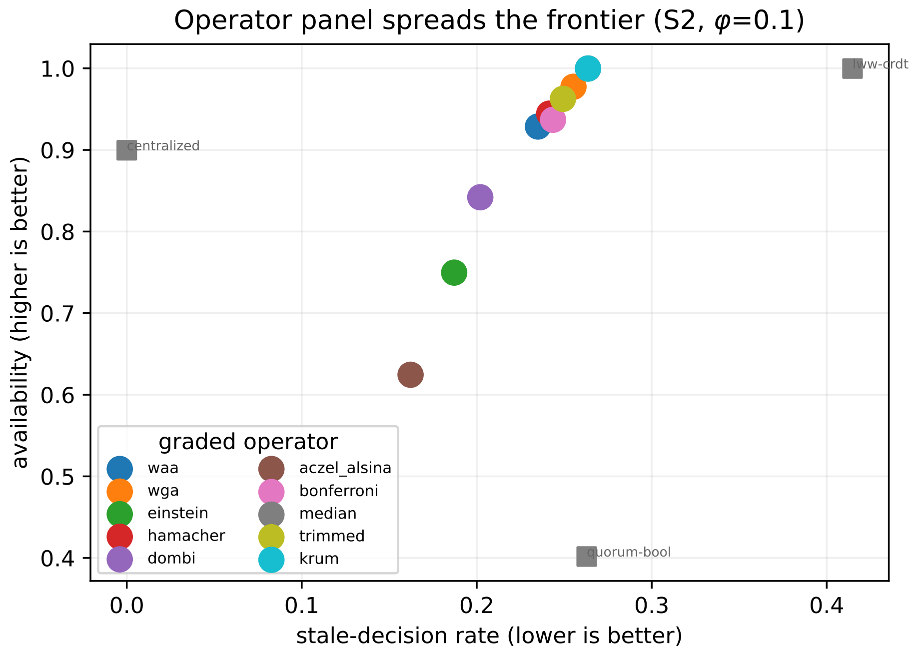

Figure 3. Axis A: the graded operator panel spreads a tunable correctness/availability frontier at fixed τ (S2, $\varphi = 0.1$), from conservative (Aczél–Alsina) to optimistic (median/Krum), bracketed by the consistency baselines. On crisp inputs every operator collapses to WAA.

kinds are the DSL: microservice logs are JSON parsed by an ANTLR grammar into a tree walked by generated listeners/visitors. Business rules (Yin et al. tag sets) are JSON parsed by the same pipeline and evaluated against facts, so a rule is data, changeable without redeploying a service.

4.3 The decentralized decision protocol

For a key under decision, each reachable peer reports its SVNN view; unreachable peers contribute the absent state $(0,0,1)$. The deciding node aggregates with Eq. (1) or (2), deneutrosophizes, and *acts* (serving the value confirmed by the persisted peers, falling back to cached holders) iff $s \geq \tau$, else *abstains*. No central node participates. This is a freshness decision, not replication: it never establishes agreement on a log; it decides whether the value a service already holds is trustworthy enough to act on.

4.4 Two-axis enrichment: operators, encoding, tuning, and adversaries

The submitted layer fixes one aggregation pair (Eqs. (1)–(2)), one deneutrosophy score, uniform weights, and a probabilistic failure model. To test whether any of these choices drives the result, each is generalised into an equal-footing panel and varied one axis at a time; the submitted configuration is the anchor and is reproduced exactly. Supplementary Fig. S9 shows the audit pipeline.

Axis A: aggregation operator. Beyond the optimistic SVNNWAA and conservative SVNNWGA, a panel of single-valued neutrosophic weighted operators is evaluated: Einstein and Hamacher t-norm/t-conorm aggregations, the Dombi[36] and Aczél–Alsina[37] operators (each with a parameter that recovers the arithmetic mean at a limit, verified as a correctness check), and the Bonferroni mean, which aggregates every ordered peer pair and so captures interrelation that a marginal average ignores. Holding the peer evidence fixed and swapping only the operator isolates the effect of the aggregation rule.

Graded encoding. On crisp inputs all t-conorm operators collapse to the same output—any persisted peer drives aggregate truth to one—so the operators differentiate only when peer views are graded. Each peer's view is therefore softened by an *observable* recency signal, its version lag relative to the most up-to-date reachable peer, requiring no oracle: a fresh persisted peer reads near $(1,0,0)$; a stale persisted peer shifts mass to falsity; a cached peer carries high indeterminacy that decays with lag; and a peer holding a lone anomalously high version is treated as a suspected dirty (uncommitted) write. The crisp encoding is retained as the conservative baseline, so the contrast also measures what graded confidence adds over a boolean status. Algorithm 2 states the encoding from the observable versions alone.

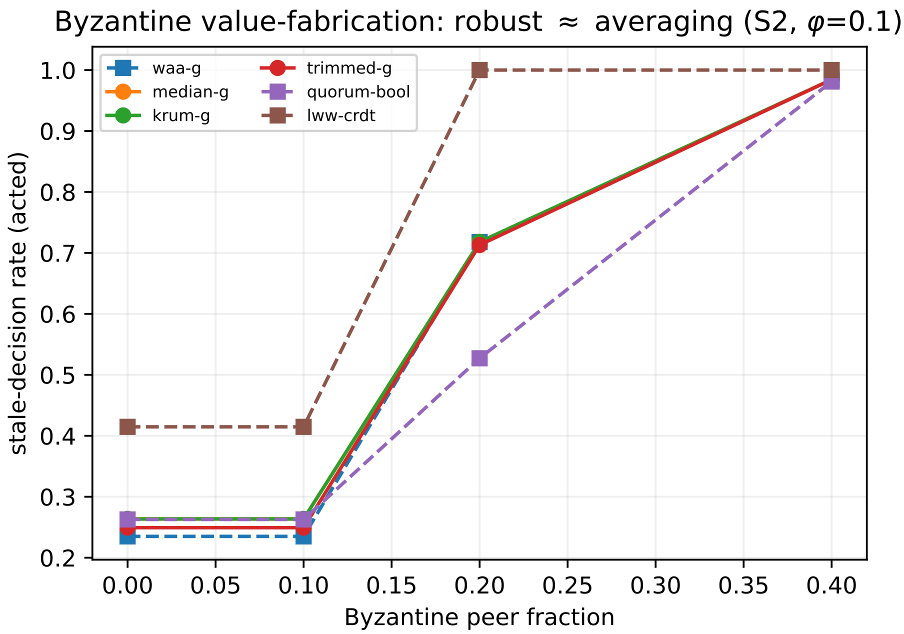

Figure 4. Byzantine value-fabrication (S2, $\varphi = 0.1$, $R = 5$): the robust aggregators (median, Krum, trimmed mean) degrade together with the averaging operators as the adversary fraction grows—robust aggregation defends the gate (Lemma 3), not value selection.

Axis A′: deneutrosophy score. The score $s(z) = (2+T-I-F)/3$[26] is one choice among several. An accuracy-degree score (truth less falsity), a cosine measure to the truth ideal, and a truth-weighted score are compared as a robustness ablation, to confirm that the freshness result does not hinge on the particular deneutrosophy.

Axis B: weight and threshold tuning. The hand-fit τ and uniform weights are replaced, in an audit arm, by tuning the full vector $[\tau, w_1, \ldots, w_R]$ against the held-out calibration utility at an *identical* evaluation budget across seven optimisers—uniform random search, differential evolution[38], CMA-ES[39], particle-swarm optimisation[40], a tree-structured Parzen estimator, the physics-based RIME optimiser[41], and the Ninja Optimisation Algorithm[42]. Because the budget, not the per-iteration mechanism, is the fair unit, the comparison asks—in the spirit of the No-Free-Lunch theorem[35]—whether the optimiser choice matters once compute is held equal. The Ninja Optimisation Algorithm has no public reference implementation and is reconstructed from its published structure with disclosed coefficients; it is a provisional panel member under test, never a hero method.

Byzantine faults and robust aggregation. The submitted failure model is probabilistic. An adversarial arm injects Byzantine peers that claim PERSISTED while holding a fabricated value at an inflated version—the worst case for an optimistic operator and for majority value selection. Three robust aggregators then replace the averaging rule: the coordinate-wise median and trimmed mean, which tolerate a strict minority of adversarial triples[43], and Krum[44], which selects the peer triple closest to its neighbours. Section 4.5 bounds their resistance at the act/abstain gate; unlike PBFT-style agreement[19], the freshness gate does not attempt cross-peer consensus on a value. A freshness-SLO reader—a per-read dual of adaptive-freshness schedulers[9] and consistency-based SLAs[13,34] that acts only when the latest value is confirmed-committed—is added so the adaptive-consistency line the layer is positioned against is compared empirically, not only in prose. Algorithm 3 states the robust gate.

Complexity. A decision issues a single query round to the R reachable peers, so the *communication* cost is $O(|R|)$ messages per decision—one round-trip per peer ($O(1)$ for the local and single-hop baselines, $O(|R|)$ for every fan-out strategy, reported exactly in Supplementary Tables S2 and S3). The *computation* per decision is $O(|R|)$ for the averaging operators (a single linear fold of the R weighted triples), with the deneutrosophy score and gate $O(1)$; the Byzantine-robust aggregators cost $O(|R| \log |R|)$ (coordinate-wise median, trimmed mean) or $O(|R|^2)$ (Krum's pairwise-distance selection). Memory is $O(|R|)$ for the collected views. The layer's per-decision cost is therefore $O(|R|)$ with the averaging operators—the same order as the quorum read it performs—and super-linear only for the optional robust aggregators.

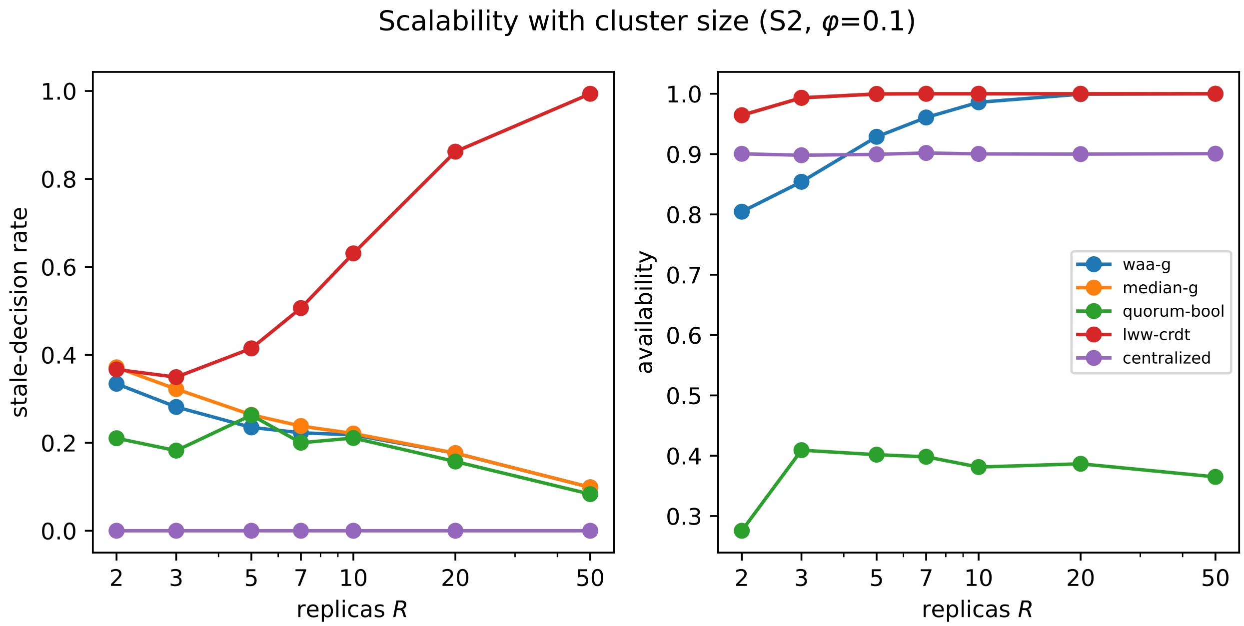

Figure 5. Scalability with replica count R (S2, $\varphi = 0.1$): the neutrosophic layer's stale-rate falls and availability rises with R (at R messages/decision), whereas eventual consistency degrades with scale and the boolean quorum stays low-availability.

4.5 Formal properties

The protocol is analysed under the crisp encoding (peers report the corners $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ for persisted, cached, absent); the graded encoding interpolates the gate continuously between the bounds below. Let R be the set of reachable peers with normalized weights w_j (the results below hold for arbitrary w_j, hence for the tuned weights of Axis B).

Proposition 1 (one-round termination, partition tolerance)**.** *Algorithm 1 issues exactly one query round and always returns* ACT *or* ABSTAIN*; it never blocks, for any pattern of failures or partitions.*

Proof. The loop visits each peer once; unreachable peers contribute $(0,0,1)$ rather than waiting, so the gate $s \geq \tau$ is a total function of the collected views. □

The layer is therefore always available in the CAP sense, trading consistency for availability: a partitioned side decides from its local view alone.

Lemma 1 (SVNNWGA conservative safety)**.** *Under SVNNWGA with $\tau \in (\frac{1}{3}, 1]$, the layer acts only if* every *reachable peer reports* PERSISTED.

Proof. $\bar{z}.T = \prod_{j \in R} T_j^{w_j} = 0$ once any reachable peer has $T_j = 0$ (cached or absent); that peer has $I_j = 1$ or $F_j = 1$, so $\bar{z}.I = 1 - \prod(1-I_j)^{w_j} = 1$ or $\bar{z}.F = 1 - \prod(1-F_j)^{w_j} = 1$, giving $s = (2 + 0 - \bar{z}.I - \bar{z}.F)/3 \leq \frac{1}{3} < \tau$. A non-unanimous round therefore abstains. □

Lemma 2 (SVNNWAA optimistic liveness)**.** *Under SVNNWAA with $\tau \leq 1$, the layer acts whenever at least one reachable peer reports* PERSISTED.

Proof. $\bar{z}.T = 1 - \prod_{j \in R}(1-T_j)^{w_j} = 1$ once any $T_j = 1$; that persisted peer has $I_j = F_j = 0$, so $\bar{z}.I = \prod I_j^{w_j} = 0$ and $\bar{z}.F = \prod F_j^{w_j} = 0$, whence $s = 1 \geq \tau$. □

Lemmas 1–2 bracket the gate for the *crisp* encoding under arbitrary weights w_j: every panel operator (Einstein, Hamacher, Dombi, Aczél–Alsina, Bonferroni) acts on a condition between "some persisted" (SVNNWAA) and "all persisted" (SVNNWGA). The next proposition lifts this to the *graded* encoding and tuned weights, answering "what does safety mean when the encoding is graded and the weights are tuned?".

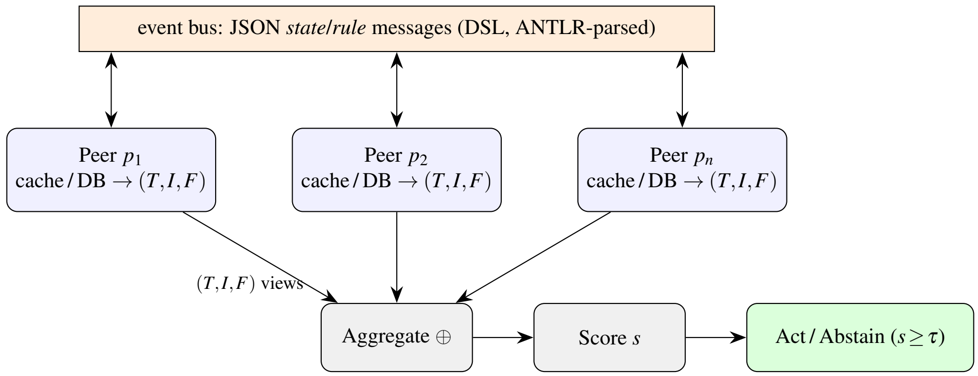

Figure 6. The decentralized neutrosophic decision layer. Sovereign peers exchange JSON state/rule messages over an event bus and tag each value's storage status as a single-valued neutrosophic number; a deciding service fuses the reachable peers' (T, I, F) views with an aggregation operator, deneutrosophizes to a score s, and acts on a value or abstains against a calibrated threshold τ—with no central authority.

Algorithm 1 Decentralized neutrosophic freshness decision for key k

Require: peers P, weights w, operator $\oplus \in \{\text{SVNNWAA}, \text{SVNNWGA}\}$, threshold τ

$Z \leftarrow \{\}$
for $p \in P$ **do**
  $z_p \leftarrow \textsc{Query}(p, k)$ **if** p reachable **else** $(0, 0, 1)$ ▷ absent reads as falsity
  $Z \leftarrow Z \cup \{(p, z_p)\}$
end for
$\bar{z} \leftarrow \oplus(\{z_p\}, w)$ ▷ Eq. (1) or (2)
$s \leftarrow (2 + \bar{z}.T - \bar{z}.I - \bar{z}.F)/3$ ▷ deneutrosophy
if $s \geq \tau$ **then**
  return $\textsc{Act}$(majority value among persisted (else cached) peers)
else
  return $\textsc{Abstain}$
end if

Proposition 2 (graded envelope and monotonicity)**.** *Let each reachable peer carry the graded triple induced by its freshness $\varphi_j = 1/(1 + \ell_j/\sigma)$ (version-lag ℓ_j, softness σ; Algorithm 2), under arbitrary normalized weights w_j. For every panel operator the deneutrosophied score s is non-decreasing in each φ_j. Consequently the acting set $\{s \geq \tau\}$ is monotone non-increasing in the threshold τ and monotone non-decreasing in the softness σ, and is contained between the SVNNWAA and SVNNWGA acting sets.*

Proof. Each operator is a weighted t-conorm/t-norm, coordinate-wise monotone in (T, I, F); the graded map sends larger φ_j to larger T_j and smaller I_j, F_j, and composition preserves monotonicity, so s is non-decreasing in every φ_j. Raising τ only removes views from the super-level set $\{s \geq \tau\}$; φ_j is increasing in σ, so increasing σ raises s and enlarges the set. The envelope holds because every weighted mean lies between its product (SVNNWGA) and co-product (SVNNWAA) aggregates. □

Corollary 1 (bounded staleness of acted values under SVNNWGA)**.** *Under SVNNWGA with the graded encoding and $\tau \in (\frac{1}{3}, 1]$, the layer acts only when every reachable peer has freshness $\varphi_j \geq \varphi_\tau$ for a threshold φ_τ fixed by τ; equivalently every acted value comes from peers whose version-lag satisfies $\ell_j \leq \sigma(1 - \varphi_\tau)/\varphi_\tau$. The staleness of any acted value is therefore bounded* a priori *by $\sigma(1 - \varphi_\tau)/\varphi_\tau$, which the operator and τ control directly.*

Proof. By Proposition 2 SVNNWGA has the smallest acting set; its geometric truth aggregate $\prod_j \varphi_j^{w_j}$ clears the gate only if every factor does, giving the per-peer floor φ_τ; invert $\varphi_j = 1/(1 + \ell_j/\sigma)$. □

Lemma 3 (Byzantine gate resistance)**.** *Let b reachable peers be Byzantine, each reporting an arbitrary triple, with $b < \lceil R/2 \rceil$ so honest peers hold a strict majority* in every coordinate. *Under a coordinate-wise median aggregate, each of $\bar{z}.T, \bar{z}.I, \bar{z}.F$ lies between two honest order statistics; the gate score equals the honest-only median, and the adversaries cannot force $s \geq \tau$ when the honest majority would abstain.*

Algorithm 2 Graded confidence encoding from observable version lag

Require: reachable peer views P with versions; lag scale σ

$v_{\max} \leftarrow \max_{p \in P} \mathrm{ver}(p)$; $\quad v_2 \leftarrow$ second-largest version in P
for $p \in P$ **do**
    $\ell \leftarrow v_{\max} - \mathrm{ver}(p)$ ▷ relative lag; no oracle
    $\phi \leftarrow 1/(1+\ell/\sigma)$ ▷ freshness: 1 at ℓ=0, $\to 0$ as ℓ grows
    $\mathit{dirty} \leftarrow [\mathrm{ver}(p) > v_2 \wedge \mathrm{status}(p) = \textsc{cached}]$ ▷ lone high version
    $z_p \leftarrow \textsc{Grade}(\mathrm{status}(p), \phi, \mathit{dirty})$ ▷ $\to (T, I, F)$, Sec. 4.4
end for
unreachable peers contribute $\textsc{Grade}(\textsc{absent}, \cdot, \cdot)$
return $\{z_p\}$

Algorithm 3 Byzantine-robust freshness gate

Require: graded views $Z = \{z_p\}$; threshold τ; rule $\rho \in \{\textsc{median}, \textsc{trimmed}, \textsc{krum}\}$; assumed adversaries f

if $\rho = \textsc{median}$ **then**
    $\bar{z} \leftarrow$ coordinate-wise median of Z
else if $\rho = \textsc{trimmed}$ **then**
    $\bar{z} \leftarrow$ coordinate-wise mean of Z after dropping the f smallest and f largest per axis
else if $\rho = \textsc{krum}$ **then**
    $\bar{z} \leftarrow \arg\min_{z \in Z} \sum$ of squared distances from z to its $|Z| - f - 2$ nearest views
end if
$s \leftarrow (2 + \bar{z}.T - \bar{z}.I - \bar{z}.F)/3$
return $\textsc{Act}$ on the persisted-majority value **if** $s \geq \tau$ **else** $\textsc{Abstain}$

Proof. With $b < \lceil R/2 \rceil$ corruptions the median of R values in each coordinate is bracketed by honest order statistics; the score is a fixed monotone function of the three coordinate medians. □

Lemma 3 protects the *gate*—the act/abstain confidence—but not the *value* returned. We state that gap precisely, as it is the layer's dominant adversarial exposure.

Proposition 3 (value selection is not Byzantine-robust)**.** *The value returned on* ACT *is the majority value among peers reporting* PERSISTED *(Algorithm 1). Coordinate-wise robustness of the gate does not constrain this choice: if $b \geq 1$ Byzantine peers report* PERSISTED *on a common fabricated value at an inflated version and the honest persisted peers do not strictly outnumber them, the fabricated value can be selected even though the gate is satisfied. Hence gate-robustness does not imply value-robustness.*

Proof. The gate aggregate is invariant to which value underlies a PERSISTED report; value selection is a separate majority over reported values, which a colluding block of size at least the honest persisted count can win. □

This matches Figure 4 (robust and averaging readers degrade together under value fabrication) and pinpoints the fix: a Byzantine-robust *value* quorum beneath the gate, which we leave to future work (Section 3).

Theorem 1 (committed-value safety, non-adversarial model)**.** *In the crash/partition (non-adversarial) model, under SVNNWGA with $\tau \in (\frac{1}{3}, 1]$ the layer never acts on a value that was never committed.*

Proof. By Lemma 1 SVNNWGA acts only on a unanimous PERSISTED round; a truthful PERSISTED report on the single-writer ground truth marks a committed value, so the served majority value is committed. □

Split-brain. By Proposition 1 each side of a partition decides independently in one round, so neither blocks. By Lemma 1 an SVNNWGA side acts only on unanimous local confirmation; with a single-writer ground truth it then serves the value its persisted peers hold, and the layer never asserts cross-partition agreement — consistent with its stated scope as a freshness gate above the consistency substrate, not a consensus protocol.

4.6 Scenarios and ground truth

Four scenarios instantiate the study. Three are e-commerce: (S1) personalized dynamic pricing (read-heavy), (S2) suspicious-order fraud detection (asymmetric cost—a missed fraud is worse than a false alarm), and (S3) "White Friday" load-spike with stateful container cloning, where a fresh replica catches up by replaying the logs-as-DSL stream. The fourth probes a *distinct* domain to test cross-domain generality: (S4) an ML *online feature store*, where features are updated frequently (high version-lag and dirty-write rates), served largely from cache, and a stale served feature degrades a downstream prediction (high stale-cost)—a stylized but timely regime, not real traces (replaying production traces remains future work, Section 3). A privileged *oracle*—written only by the workload, never by a service—records the totally ordered commits and reconstructs the true global state at any logical time. A decision is *stale* iff the value acted on differs from the oracle's true value.

4.7 Baselines

We compare eight decision strategies on identical decision streams: our **neutro-waa** and **neutro-wga**; *centralized* (a strongly-consistent authority—the correctness ceiling); *raft-lww* (a leader's committed value); *quorum-bool* (majority vote over boolean fresh/stale flags—the directly comparable control that isolates the value of (T,I,F) versus a bit); *prob-gate* (a calibrated single-probability gate that collapses the same evidence into one Bayesian freshness posterior $p=(a+1)/(a+b+2)$ and acts iff $p \geq \tau$—the sharper control that isolates (T,I,F) vs. a single scalar); *pbs-quorum* (a probabilistically-bounded-staleness partial-quorum reader[13], the nearest prior art); *lww-crdt* (eventually consistent last-writer-wins); *single-peer*; and *naive-cache* (local read, no coordination—the floor).

4.8 Experimental design and statistics

Each cell is $(\text{scenario}, \text{system}, \varphi, \text{partition})$ at $R=5$ replicas, cache ratio $\rho=0.5$, with failure-injection rate $\varphi \in \{0, 0.1, 0.2, 0.3\}$ and partition $\in \{\text{none}, \text{transient}\}$; the workload models clean-cached, dirty-cached (an uncommitted later write—the dirty-read phenomenon), persisted-latest, confidently-stale-persisted, and absent peers. Each cell runs 50 repetitions of 300 decisions under five master seeds—250 repetitions per cell. The environment RNG excludes the system name, so every strategy sees the *identical* decision and failure stream (a paired design). *Unit of analysis.* Each repetition contributes one summary statistic (its mean stale-rate or availability over its 300 decisions), and every significance test pairs strategies *at the repetition level*: the sample size of each paired test is the number of repetitions ($n=250$ per cell), not the number of individual decisions, so within-repetition correlation among decisions does not inflate the effective n. The acceptance threshold τ is fit per (scenario, operator) on *held-out* calibration seeds by maximizing availability minus a per-scenario cost on the stale-rate, then frozen. A single canonical aggregator emits all tables and figures; it asserts a common calibration hash across inputs and refuses to mix files that disagree (R6). For full transparency the locked calibration's canonical-JSON SHA-256 is `16dc5ef33908a3723eb896b1bba1129ed147643f8a3bef4a55c459f9ec836070`. The aggregator computes stale-rate over acted decisions (success-only) and availability separately, applies Holm correction within a-priori comparison families, reports paired Cohen's d with 5000-resample bootstrap 95% CIs, and flags any cell whose headline contrast does not replicate in sign across all five seeds.

Correctness and availability come from this discrete-event simulation, which scales to many cells; *decision latency* is measured separately on a real testbed: each peer is a live ASGI endpoint and the deciding node drives it with a real asynchronous HTTP client, so every strategy executes its true communication pattern (one hop, wait-for-majority, or wait-for-all plus real SVNN aggregation) over real sockets with a configurable per-hop RTT. A Docker Compose deployment of the same topology (per-service containers plus a Redis event bus) is provided for fidelity and reuse.

Data availability

The aggregated result summaries underlying every figure and table in this article—the per-configuration outcome summaries, the pairwise statistical comparisons, and the multi-seed replication checks—are provided with this article as Supplementary Information. All results are generated by the evaluation simulator under a fixed, content-hashed calibration and the recorded random seeds described in the Methods; together with the figure and table generators they regenerate every reported number. The complete raw per-trial outputs, together with the implementation, are archived in a single Zenodo record (DOI: 10.5281/zenodo.20932989). During peer review this record is held under restricted access to protect ongoing work; editors and reviewers are granted immediate access through a private repository link provided to the editorial office, and it will be released openly upon publication (data under the Creative Commons Attribution 4.0 International licence, code under the MIT licence).

Code availability

The implementation (neutrosophic operators, the ANTLR-based DSL and rules engine, the decision protocol, the baselines, and the evaluation harness) is archived in the same Zenodo record as the data (DOI: 10.5281/zenodo.20932989). During peer review it is held under restricted access; editors and reviewers are granted immediate access through a private repository link provided to the editorial office, and it will be released publicly under the MIT licence upon publication. Dependency versions are pinned and the generated parser is included, so no Java toolchain is needed to reproduce the results.

Author contributions

H.A.E. conceived and designed the study, implemented the system, performed the experiments, analysed the data, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Use of large language models

A large language model (Claude, Anthropic) assisted with code scaffolding and manuscript drafting under the author's direction; the author reviewed and verified all content and is solely responsible for it.

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